Foundations of Generative AI (ICT4403)

Assignment 2: Autoregressive Text Generation System Submission

1. Submission & Repository Identification

Student Name	Rusheel Reddy
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Assignment Title	Assignment 2: Comparative Autoregressive Text Generation
System Title	Autoregressive Next-Word Generation System using Simple RNN, Bidirectional RNN, LSTM, and GRU
Dataset Name & Source Link	The Adventures of Sherlock Holmes (Sir Arthur Conan Doyle) Project Gutenberg: https://www.gutenberg.org/ebooks/1661
GitHub Repository Link	https://github.com/RK-0111/Fgai-assignment

2. Executive System Overview

This project develops, trains, and empirically compares an autoregressive text generation system formulated for next-word prediction across four distinct recurrent neural network architectures: Simple RNN, Bidirectional RNN, LSTM, and GRU. All four models were trained under strictly comparable hyperparameter budgets (embedding dimension 128, hidden dimension 128, 2 recurrent layers, Adam optimizer, lr = 0.002, Cross-Entropy Loss, and sequence length 12) on a curated corpus of Sir Arthur Conan Doyle's 'The Adventures of Sherlock Holmes' from Project Gutenberg.

3. Quantitative Performance Summary

Model	Parameters	Train Loss	Val Loss	Perplexity	Top-1 Acc	Time (s)
Simple RNN	708,548	4.5351	5.1805	177.78	15.01%	51.13s
Bidirectional RNN	692,164	4.4007	5.2302	186.84	15.32%	86.17s
LSTM	906,692	4.6602	5.1230	167.84	15.17%	94.69s
GRU	840,644	4.4772	5.0970	163.53	16.32%	146.44s

4. Key Findings & Best-Performing Architecture

- Optimal Model Selection: The Gated Recurrent Unit (GRU) achieved the lowest validation loss (5.0970), lowest validation perplexity (163.53), and highest top-1 accuracy (16.32%), providing superior autoregressive modeling quality.
- Computational Trade-offs: Simple RNN achieved the fastest convergence (51.13s) due to lack of gating logic, but suffered from higher perplexity. GRU balanced gating sophistication with fewer parameters (840k) compared to LSTM (906k).

- Temperature Sampling Dynamics: At T = 0.3, generations were deterministic with n-gram repetition loops. At T = 0.7, all models attained optimal lexical diversity (Distinct-1 ~0.77) and syntactical coherence. At T = 1.2, high entropy introduced grammatical degradation.
- Deliverables Provided: The full GitHub repository includes source code (.py), an interactive Jupyter Notebook (.ipynb), high-resolution metric figures, generated text JSON dumps, and a comprehensive README.